

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	DSA0307	Course Title:	Natural Language Processing
CO Assessed:	CO1 – Imitation (L1)	Assessment:	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA		
CO1 Statement:	Analyse NLP models, regular expression patterns, finite state automata, morphological processing techniques and to interpret language processing. (BL-3)		
Student Name:	T Deekshith	Reg. No.:	192472253
Section / Batch:	CSE-AI , 2024	Date:	16/07/26

Code of Conduct

I ___T Deekshith - 192472253___ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: T Deekshith

Section / Topic	Focus	Q Nos
Information Extraction	Regex-Based Information Extraction	1
Pattern Matching	Regex Pattern Matching	2
Regular Expression Patterns	Regex-Based Input Validation	3

Experiment 1: Resume Information Extraction Using Regular Expressions

Aim:

To develop a Python program using Regular Expressions to extract candidate information from a resume.

Algorithm:

1. Import the `re` module.
2. Store the resume as a string.
3. Extract name, email, mobile number, skills, and experience using Regex.
4. Display the extracted information.
5. Check eligibility (Experience ≥ 2 years and Python skill).
6. Display the result.

Program

```
import re

resume = """
Name: Rahul Sharma
Email: rahul@gmail.com
Mobile: 9876543210
Skills: Python, Java, SQL, Machine Learning
Experience: 3 years
"""

name = re.search(r"Name:\s*(.*)", resume)
email = re.search(r"[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}", resume)
mobile = re.search(r"[6-9]\d{9}", resume)
skills = re.findall(r"Python|Java|SQL|Machine Learning|NLP", resume)
experience = re.search(r"(\d+)\s+years", resume)

print("Name :", name.group(1))
print("Email :", email.group())
print("Mobile :", mobile.group())
print("Skills :", skills)
print("Experience :", experience.group(1), "Years")

if int(experience.group(1)) >= 2 and "Python" in skills:
    print("Eligible for Shortlisting")
else:
    print("Not Eligible")
```

Sample Output

```
Name : Rahul Sharma
Email : rahul@gmail.com
Mobile : 9876543210
Skills : ['Python', 'Java', 'SQL', 'Machine Learning']
Experience : 3 Years
Eligible for Shortlisting
```

Result:

Thus, the resume information was extracted successfully using Regular Expressions.

Experiment 2: Product Search System Using Regular Expressions

Aim:

To develop a Python program for searching products using Regular Expressions.

Algorithm:

1. Import the `re` module.
2. Store product names in a list.
3. Get the search keyword.
4. Perform exact, prefix, suffix, partial, and case-insensitive search.
5. Display matching products.
6. Display total number of matches.

Program

```
import re

products = [
    "Laptop",
    "Laptop Bag",
    "Gaming Laptop",
    "Mouse",
    "Wireless Mouse",
    "Keyboard",
    "Smartphone"
]

key = input("Enter Product Keyword: ")

count = 0

print("\nMatching Products:")

for product in products:
    if re.search(key, product, re.IGNORECASE):
        print(product)
        count += 1

print("\nExact Match")
for product in products:
    if re.fullmatch(key, product, re.IGNORECASE):
        print(product)

print("\nPrefix Match")
for product in products:
    if re.match(key, product, re.IGNORECASE):
        print(product)

print("\nSuffix Match")
for product in products:
    if re.search(key + "$", product, re.IGNORECASE):
        print(product)

print("\nTotal Matches =", count)
```

Sample Output

Enter Product Keyword: Laptop

Matching Products:

Laptop
Laptop Bag
Gaming Laptop

Exact Match

Laptop

Prefix Match

Laptop
Laptop Bag

Suffix Match

Laptop
Gaming Laptop

Total Matches = 3

Result:

Thus, product searching was successfully implemented using Regular Expressions.

Experiment 3: University Registration Validation Using Regular Expressions

Aim:

To validate student registration details using Regular Expressions.

Algorithm:

1. Import the `re` module.
2. Read student details.
3. Validate Register Number.
4. Validate Email ID.
5. Validate Course Code.
6. Validate Semester.
7. Validate Mobile Number.
8. Display validation status.
9. Display registration result.

Program

```
import re

reg = input("Enter Register Number: ")
email = input("Enter Institutional Email: ")
course = input("Enter Course Code: ")
sem = input("Enter Semester: ")
mobile = input("Enter Mobile Number: ")

if re.fullmatch(r"\d{2}[A-Z]{3}\d{4}", reg):
    print("Register Number : Valid")
else:
    print("Register Number : Invalid")
```

```

if re.fullmatch(r"[A-Za-z0-9._%+-]+@saveetha\.com", email):
    print("Email : Valid")
else:
    print("Email : Invalid")

if re.fullmatch(r"[A-Z]{3}\d{2}", course):
    print("Course Code : Valid")
else:
    print("Course Code : Invalid")

if re.fullmatch(r"[1-8]", sem):
    print("Semester : Valid")
else:
    print("Semester : Invalid")

if re.fullmatch(r"[6-9]\d{9}", mobile):
    print("Mobile Number : Valid")
else:
    print("Mobile Number : Invalid")

if (re.fullmatch(r"\d{2}[A-Z]{3}\d{4}", reg) and
    re.fullmatch(r"[A-Za-z0-9._%+-]+@saveetha\.com", email) and
    re.fullmatch(r"[A-Z]{3}\d{2}", course) and
    re.fullmatch(r"[1-8]", sem) and
    re.fullmatch(r"[6-9]\d{9}", mobile)):
    print("\nRegistration Successful")
else:
    print("\nRegistration Failed")

```

Sample Output

```

Enter Register Number: 23ADS1023
Enter Institutional Email: student@saveetha.com
Enter Course Code: NLP03
Enter Semester: 5
Enter Mobile Number: 9876543210

```

```

Register Number : Valid
Email : Valid
Course Code : Valid
Semester : Valid
Mobile Number : Valid

```

```

Registration Successful

```

Result:

Thus, the student registration details were successfully validated using Regular Expressions.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Q. No. / Criteria	Understanding of Problem Context	Application of Concepts	Problem-Solving Approach	Accuracy of Solution	Clarity	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____